PUBLICATION CLUSTERS

P R O C E S S   B O O K

Mapping the invisible flow of scientific knowledge across two decades

Three questions this project answers:

Q1 Is the global academic network integrating or fragmenting over time?

Q2 Which countries transitioned from knowledge consumers to exporters?

Q3 Which institution pairs share the strongest reciprocal citation bonds?

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Website: com-480-data-visualization.github.io/publication-clusters/

§ 1. PROJECT OVERVIEW

Where Does Science Come From?

Every time a researcher publishes a paper, they cite the work they built on. Those citations are more than footnotes, they are a map of how ideas travel across the world. We made that map easy to read!

Publication Clusters makes it visible. We turn twenty years of citation data into a live 3D globe: each dot is a research hub, each arc is a citation link, and a year slider lets you watch the geography of scientific influence shift from 2005 to 2025 in real time. Hit play and watch two decades of knowledge flow animate automatically. Click any institution to read more about their citation history. A live leaderboard tracks the top knowledge producers and consumers at any point in time.

We cover five research domains and over a thousands of institutions worldwide. Spin the globe, pick a field, and see where science really comes from.

1.1 THE DATASET

Our data comes from the OpenAlex API, an open index of the world's research literature. For each of the five research domains, we collect the top 1,000 most-cited papers per year, covering 2005 to 2025. From those papers we extract citation links between institutions, geocode each university to place it on the globe, and pre-compute the network for fast loading. The result: 25,000+ institutions and over 3 million citation links, all navigable in real time.

Research domains:

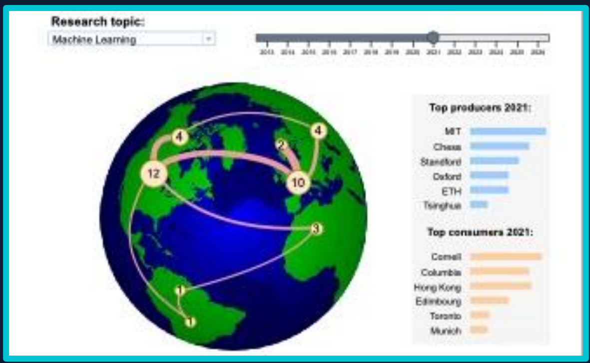
Solar Cells	LHC Physics	Complex Networks	Gravitational Waves	Magnetic Thin Films

1.2 TARGET AUDIENCE

This tool is for anyone curious about how science moves across the world, whether you are a researcher tracking your field, a university strategist or policy-maker weighing funding and collaboration decisions, or simply someone who has never thought about where ideas come from and wants to find out.

✦ 2. FROM SKETCH TO GLOBE

The Milestone 1&2 sketches showed numbered nodes on a hand-drawn globe. By Milestone 3, those became glowing CesiumJS nodes over real satellite terrain.



Initial sketch of the citation map

Final implementation of the product

> Every numbered circle in the sketch became a variable-radius glowing node. The coral arcs became geodesic great-circle paths over real satellite imagery.

2.1 WHAT CHANGED AND WHY

Globe renderer - Switched from Three.js/Globe.gl to CesiumJS for physically accurate geodesic arcs, real Bing satellite imagery, terrain depth and atmospheric haze.
Node clustering - A dynamic cell-size algorithm merges spatially close institutions into super-nodes that dissolve on zoom-in, keeping the globe readable at any altitude.
Manifest system - Hard-coded topics became a manifest.json registry. Adding a domain now requires only dropping in CSV files and updating the manifest.
Pre-computed pipeline - Live API calls were replaced by offline batch processing: OpenAlex to geocoding to static CSV files served on GitHub Pages, fully serverless.
Play button - A setInterval loop advances the year slider automatically every 900ms, toggling between play and pause. The globe animates through all years without user input.
Click-to-inspect - Clicking any node or cluster opens a side panel showing the institution name, country, total citation volume, and the breakdown between outgoing and incoming citations. For clusters, the panel lists every member institution sorted by citation weight.
Leaderboard - Each year displays the top 5 knowledge producers and top 5 knowledge consumers, loaded from pre-computed top_sources and top_targets CSV files and updated synchronously with the year slider.
Directed edges - Citation arcs use PolylineArrowMaterialProperty so the direction of knowledge flow is visually explicit, with arrow heads pointing from citing institution to cited institution.
Intro overlay - A splash screen delays data loading until the user clicks Start, keeping the initial page load fast and giving a cleaner first impression.

> All pivots share one goal: make the globe fast, honest about geography and easy to extend.

◎ 3.1 INTERACTIVE 3D GLOBE

CesiumJS was chosen because geodesic arcs, real satellite terrain and atmospheric scattering make the globe feel honest about the planet it represents.

A rotatable 3D globe powered by CesiumJS. Unlike any flat projection, a sphere provides accurate great-circle distances. Citation arcs follow true geodesic paths: the visual arc length is proportional to how far academic knowledge physically traveled.

3.1.1 VISUAL ENCODING

Node radius Citation volume: bigger node means more cited institution

Node colour Yellow for clusters of nearby institutions, pink for individual institutions

Arc origin Source institution: the citing lab

Arc endpoint Target institution: the cited lab

Arc width and opacity Citation strength: thicker and more opaque arcs represent stronger citation flows

Arrow heads Arrows point from the citing institution to the cited institution

Slider position Selected year: 2005 through 2025

Super-node A cluster of nearby institutions merged for readability; dissolves on zoom-in

> Node size encodes citation volume. Arc direction encodes knowledge flow. Arrow heads make that direction unambiguous. Year slider encodes time.

3.1.2 SUPERPOINT CLUSTERING

Raw data has thousands of overlapping nodes at world view. A dynamic cell-size algorithm merges spatially close institutions into super-nodes. Cell size adapts to camera altitude: zoomed out means large cells and few nodes; zooming in dissolves clusters back into individual institutions.

- Each super-node radius equals the sum of its constituent institutions' citation weights
- Clicking a super-node opens a detail panel listing all member institutions sorted by citation weight.
- requestRenderMode: true means the GPU only renders on state change.

◈ 3.2 LEADERBOARD PANEL

As you scrub the year slider over Solar Cells, you can watch Tsinghua climb from nowhere to permanent first place. That is the story the leaderboard tells.

A live-updating side panel shows the top-5 institutions ranked by outgoing citations (producers = knowledge exporters) and incoming citations (consumers = knowledge importers). Each entry has a proportional bar. The panel refreshes synchronously with the year slider.

3.2.2 UI CONTROLS

- Year Slider (2005 to 2025): globe, arcs and leaderboard all update simultaneously from pre-computed CSV slices
- Topic Selector: populated from manifest.json; switches the active domain with no backend call
- Play Button: a setInterval loop advances the year automatically every 900ms. Clicking again pauses.
- Click-to-inspect: clicking any node or cluster opens a side panel with the institution name, country, and a breakdown of total, outgoing and incoming citations. For clusters, the panel lists every member institution sorted by citation weight. The panel stays open as the year changes and updates its content to reflect the new year
- Live Stats Dashboard: node, edge, and cluster count and a status string describing the current view
- Superpoint Tooltips: hovering a merged node reveals all constituent institutions and combined citation weight

> We dropped animated citation icons because arc opacity already communicates citation volume. Adding particles would add noise without adding information."

⊕ 4. DATA PIPELINE

The pipeline turns raw OpenAlex API responses into lightweight static CSVs on GitHub Pages. Fully serverless.

Every visualization interaction happens client-side using pre-joined CSV slices and JavaScript Maps. There is no backend, no server, no API key exposed to the user.

4.1 FIVE STAGES

1. **OpenAlex API** Cursor-based fetch of the top 1,000 cited papers per year per topic
2. **Institution Extraction** Lead-author's institution is identified from each paper's authorship field
3. **Nominatim Geocoding** University names mapped to lat/lng via a two-pass strategy: OpenAlex then OpenStreetMap fallback
4. **Graph Construction** Edges built between institution pairs; fractional weighting applied to reflect citation strength
5. **CSV Export** geo_dict and network_evolution CSVs written to docs/data/ for static hosting

4.2 CHALLENGES

Missing GPS Coordinates About 15% of institutions in OpenAlex lack lat/lng. The two-pass strategy brought coverage to over 97%.

API Rate Limiting OpenAlex enforces strict cursor-based pagination. We added 0.1s automatic back-off per request and batched geocoding at 50 institutions per call.

Self-Loops and Noise Papers from one lab citing the same lab create noisy self-edges. Filtered client-side via HIDE_SELF_LOOPS.

File Size on GitHub Pages Domain CSVs totalled 5 to 10 MB each. Optimised by storing only source_id, target_id, weight and year.

Node and Arc Height Mismatch Nodes are rendered at 160,000m altitude above the globe surface while arcs are drawn at 20,000m. This means arcs do not visually originate from the node centres. We spent a significant amount of time trying to resolve this, experimenting with matching altitudes, adjusting arc elevation, and modifying the polyline rendering, but every attempt made more trouble. We ultimately decided to keep the current behaviour. The visual separation is subtle at most zoom levels, the directional information is preserved, and the alternative solutions created more problems than they solved."

★ 5. DELIVERY + FINDINGS

Most of the planned features in milestone 2 are now delivered. The dropped features were all optional extras: the core tool is complete and polished.

Feature	Status	Feature	Status
Rotatable 3D globe + institution nodes	Delivered	Animated citation icons	Dropped
Directed arc overlays by year	Delivered	Reciprocal citation dyad click	Dropped
Year slider 2005 to 2025	Delivered	Louvain community detection	Dropped
Top-5 producers and consumers panel	Delivered	Network density timeline	Dropped
Data pipeline OpenAlex to CSV	Delivered		
Topic dropdown filter	Delivered		
Auto-advancing animation	Delivered		

5.1 KEY FINDINGS

Q1 - Is the global academic network integrating or fragmenting over time?

We measure integration in two ways: the share of citation links that cross national borders each year, and the number of countries actively participating in each domain. Across all four domains, the total number of citation links grew roughly threefold between 2005 and 2025, and the number of countries involved rose from around 65 to over 110. The cross-border ratio has remained high and broadly stable, hovering between 80% and 84% throughout the period, with a modest upward trend peaking around 2022 before a slight retreat. The picture is one of steady integration: the network is getting larger and more globally connected, not more isolated. Individual domains tell slightly different stories - LHC saw mild domestic concentration as Chinese institutions formed tighter internal clusters, while Gravitational Waves became more internationally connected after the LIGO discovery drew in labs from across the world - but the overall trend is clear. Science is becoming more global.

Q2 - Which countries transitioned from knowledge consumers to exporters?

We measure each institution's role by subtracting its outgoing citations from its incoming citations each year. The most striking shift in our data is China's rise across multiple fields. In Complex Networks, institutions like the University of Electronic Science and Technology of China, Tsinghua University, and Shanghai Jiao Tong University all went from net importers of knowledge in 2005 to major exporters by 2025. Chinese Academy of Sciences alone swung from a score of -408 to +76 in LHC-related research. France tells a different story: CNRS and INRIA consolidated into dominant producers in Complex Networks and LHC, while French institutions in Gravitational Waves built some of the strongest domestic citation networks on the continent. Singapore's National University emerged as a significant producer in Magnetic Thin Films, shifting from -153 to +644. Going the other direction, several historically prominent institutions lost ground: the University of Porto dropped from +377 to -73 in Complex Networks, Tohoku University went from +248 to -96 in Magnetic Thin Films, and the Max Planck Institute for Gravitational Physics - once a slight producer - became one of the heaviest consumers in our dataset by 2025, reflecting how centrally it now draws on the global literature.

Q3 - Which institution pairs share the strongest reciprocal citation bonds?

We measure reciprocal bonds by taking the minimum of the citation flow in each direction between two institutions in a given year. The single strongest and most enduring bond in our entire dataset is between the California Institute of Technology (US) and the Max Planck Institute for Gravitational Physics (DE). Present in all 21 years of our data, with an average weight of 125 and a peak of 285 in 2021 following the surge of LIGO-related publications, it remained the top pair in 2025 with a weight of 139. In Complex Network Analysis, the dominant pairs shifted over time: early years were led by US-Israeli academic ties such as MIT and Weizmann Institute, but by 2018 Chinese domestic pairs had taken over, with the strongest single bond being Southwest University and the University of Electronic Science and Technology of China reaching a weight of 89 that year. The most stable domestic bond overall belongs to France in Magnetic Thin Films: CNRS and CEA have appeared together for 19 consecutive years with an average weight of 91, peaking at 152, reflecting a tightly integrated national research infrastructure that has only grown stronger over time.

◆ 6. PEER ASSESSMENT

All four members contributed equally. Responsibilities were split by component but everyone worked together and reviewed each other's work.

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- ¼ of Milestone 1 authorship
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- Data analysis
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